

Generalization Error

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Evaluation and generalization

Why minimizing the cost function (based on only training data) while we are interested in the performance on new examples?

$$\min_{\theta} \sum_{i=1}^n \text{Loss} \left(y^{(i)}, f(\mathbf{x}^{(i)}; \theta) \right) \longrightarrow \text{Empirical loss}$$

Evaluation: After training, we need to measure how well the learned prediction function can predicts the target for unseen examples

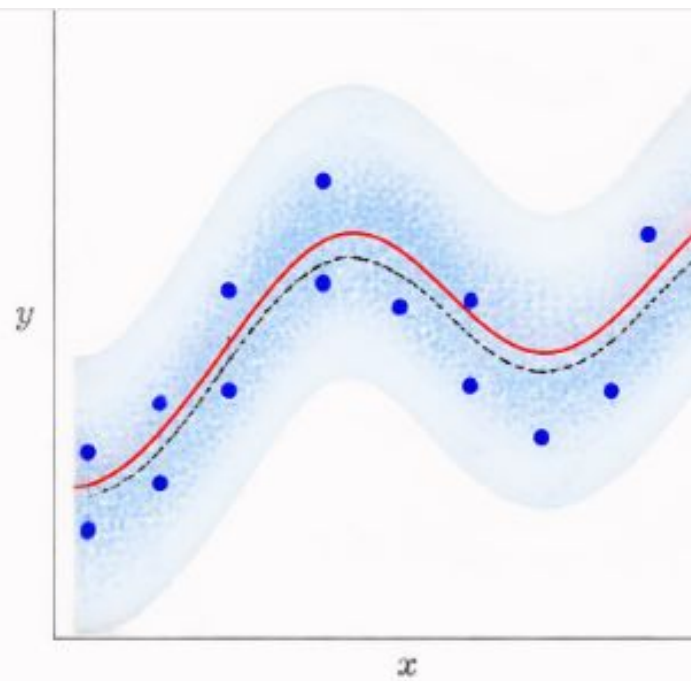
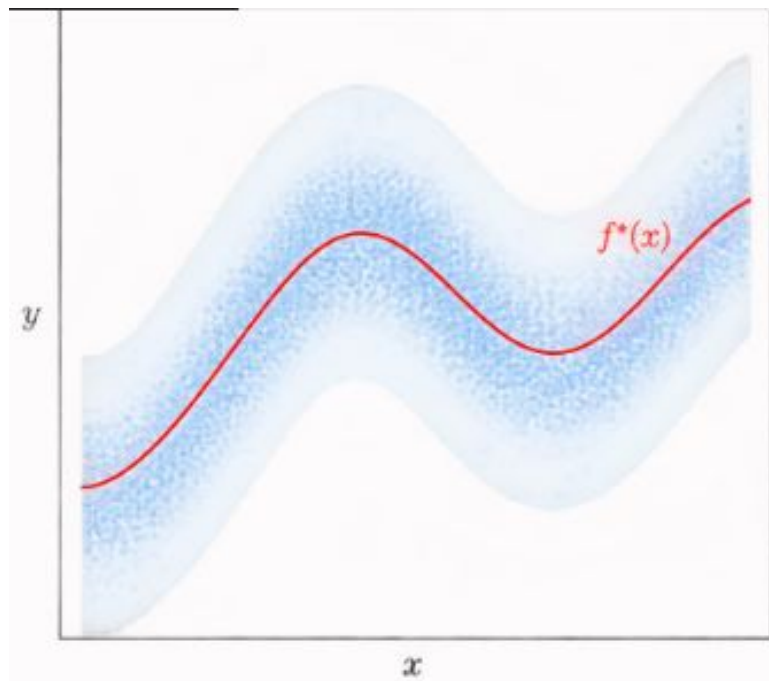
Training and test performance

Assumption: training and test examples are drawn independently at random from the same but unknown distribution.

- ▶ Each training/test example $(\mathbf{x}, y)$ is a sample from joint probability distribution $P(\mathbf{x}, y)$, i.e., $(\mathbf{x}, y) \sim P$

$$\textbf{Empirical (training) loss} = \frac{1}{n} \sum_{i=1}^n \text{Loss} \left(y^{(i)}, f(\mathbf{x}^{(i)}; \boldsymbol{\theta}) \right)$$

$$\textbf{Expected (test) loss} = E_{\mathbf{x}, y} \{ \text{Loss}(y, f(\mathbf{x}; \boldsymbol{\theta})) \}$$



Challenge of computing test (true) error

We don't know the true distribution $P(x,y)$

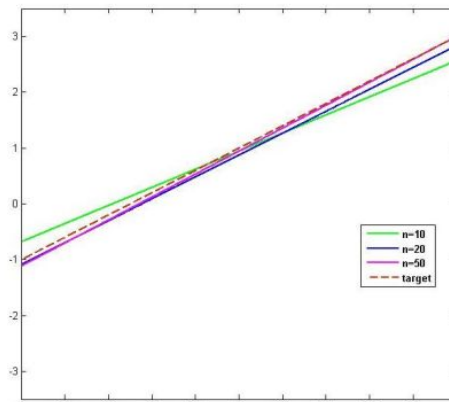
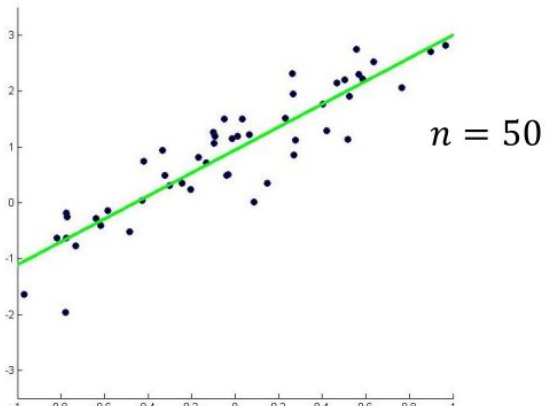
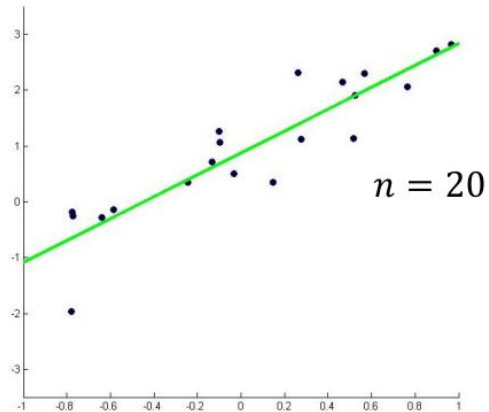
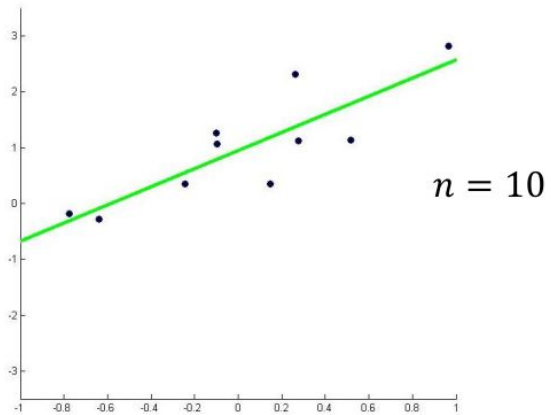
We can not calculate the exact expected test error

How do we estimate it in practice ?

We use an independent test dataset and compute the error for this dataset

Under the assumption that test samples are independently drawn from the same distribution

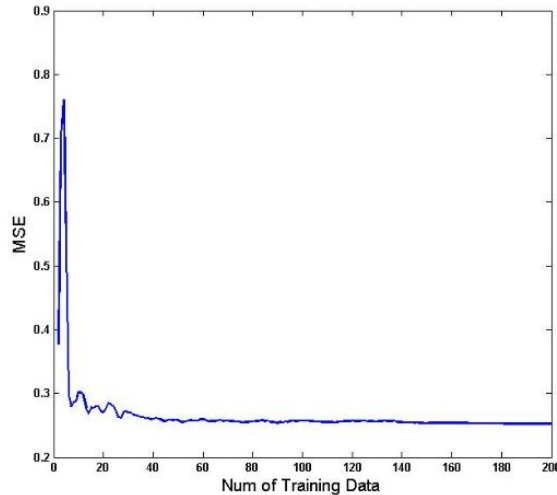
Number of training data



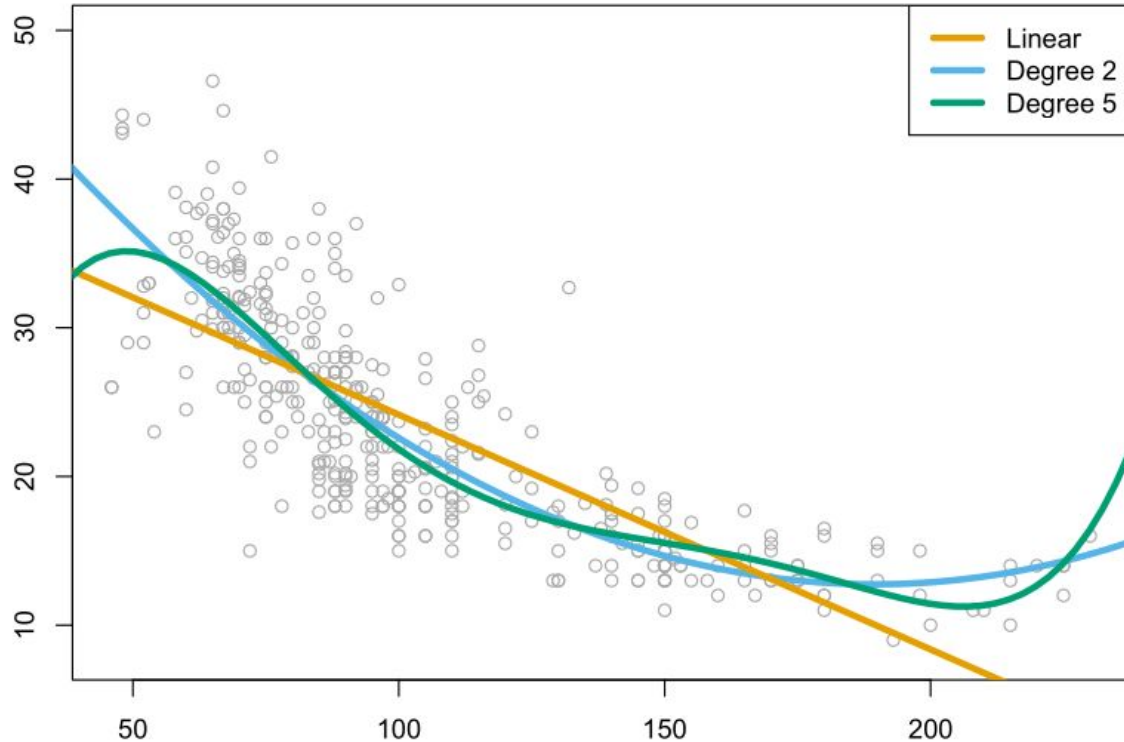
Generalization

By increasing the number of training examples, will solution be better?

Why the mean squared error does not decrease more after reaching a level?



Using evaluation for Model selection



Model complexity and overfitting

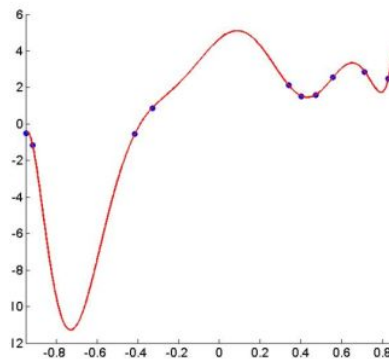
- ▶ With limited training data, models may achieve zero training error but a large test error.

Training
(empirical) loss

$$\frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - f(x^{(i)}; \theta) \right)^2 \approx 0$$

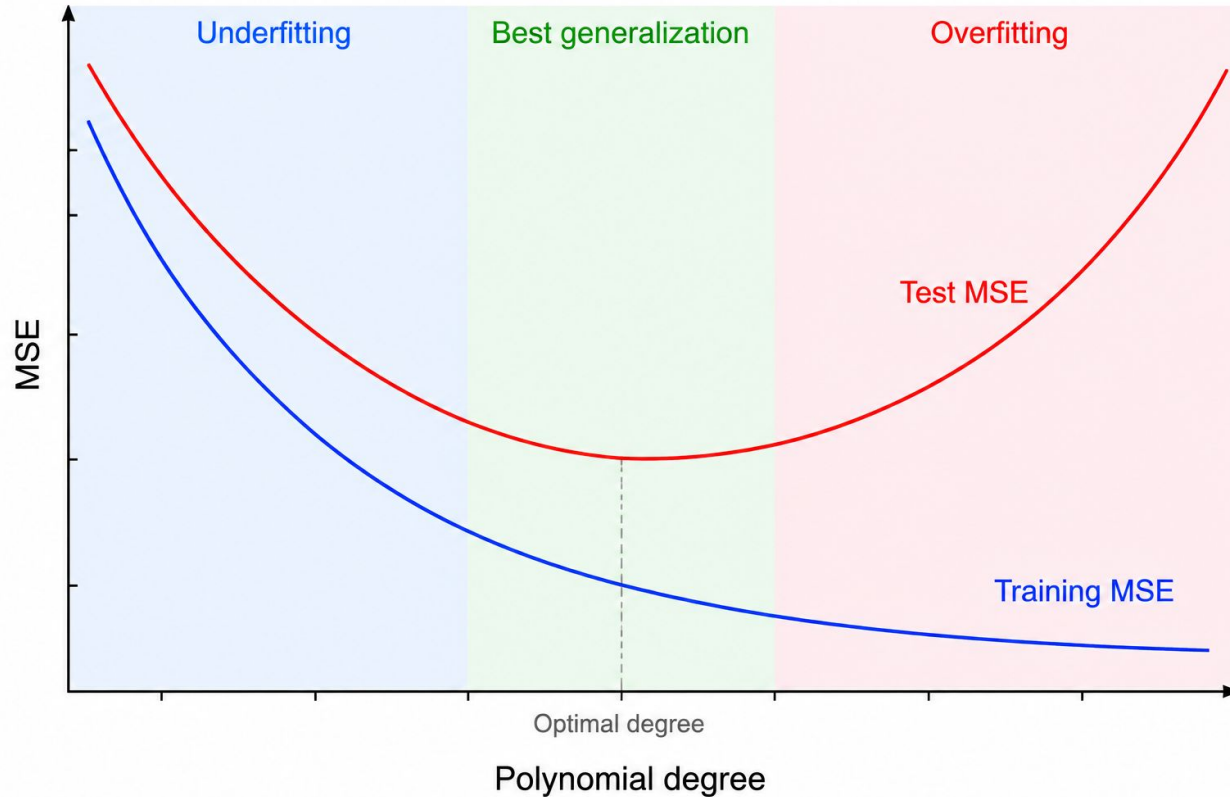
Expected
(test) loss

$$E_{\mathbf{x}, \mathbf{y}} \left\{ (y - f(\mathbf{x}; \theta))^2 \right\} \gg 0$$

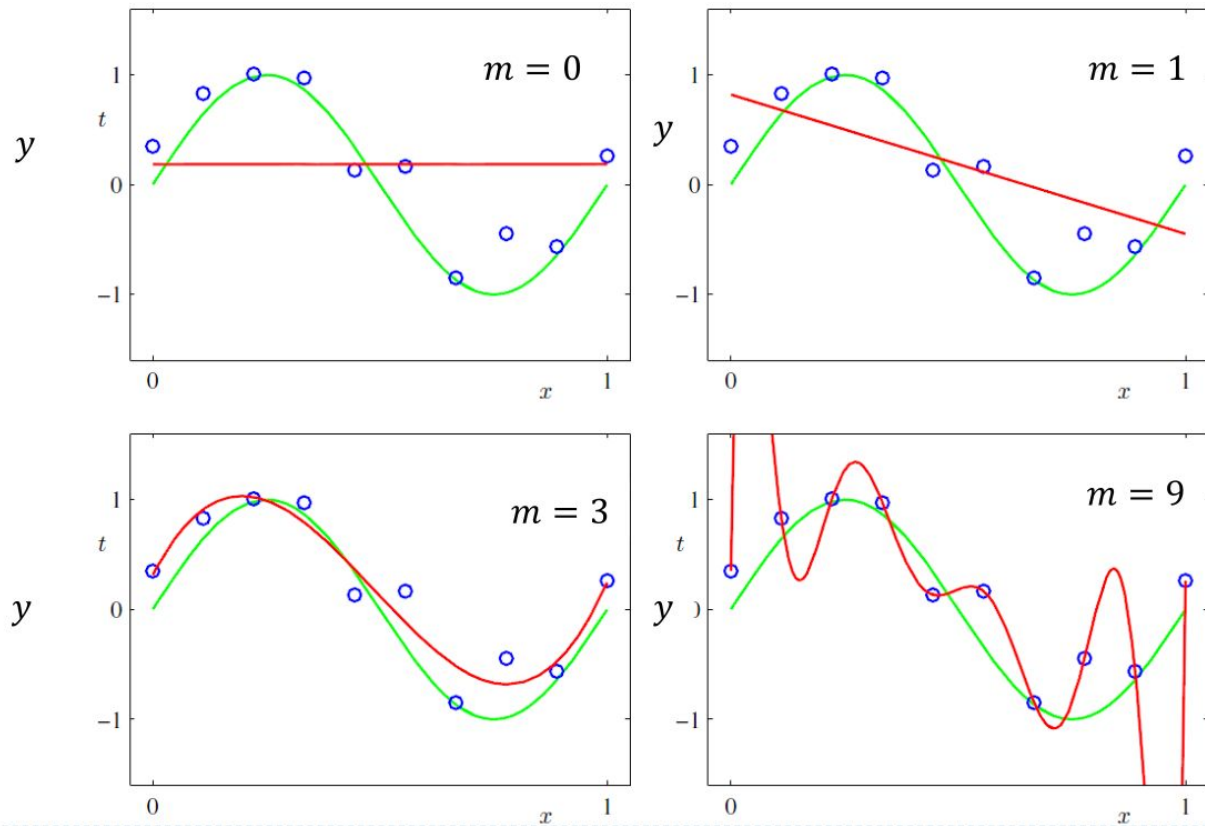


- ▶ Over-fitting: when the training loss no longer bears any relation to the test (generalization) loss.
 - ▶ Fails to generalize to unseen examples.

Polynomial regression (train and test error)



Polynomial regression (model complexity)



Overfitting causes

Model complexity

- ▶ E.g., Model with a large number of parameters (degrees of freedom)

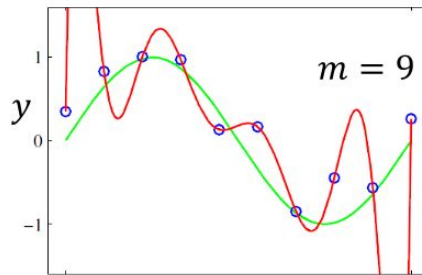
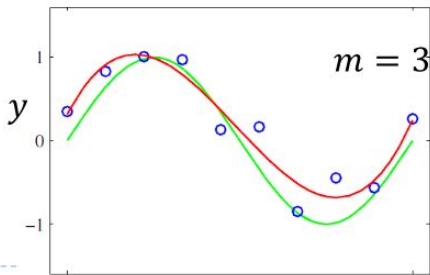
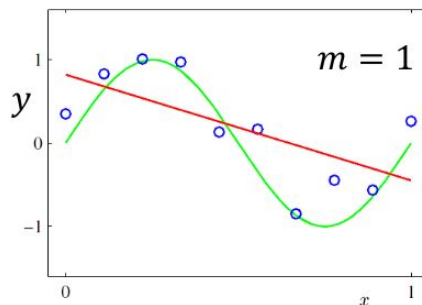
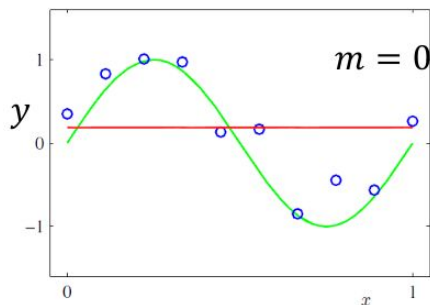
Low number of training data

- ▶ Small data size compared to the complexity of the model

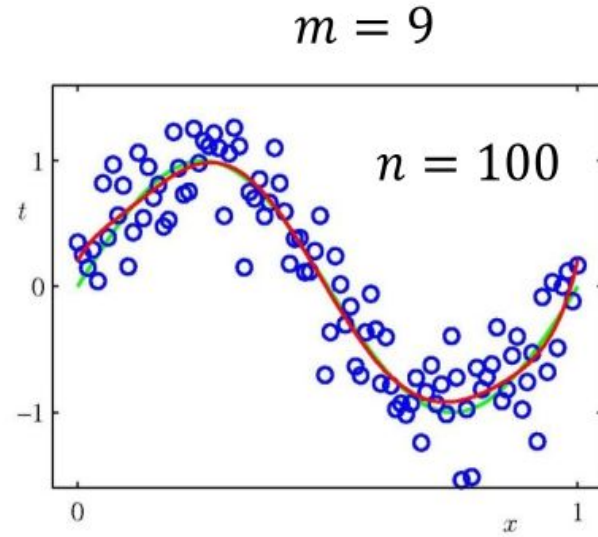
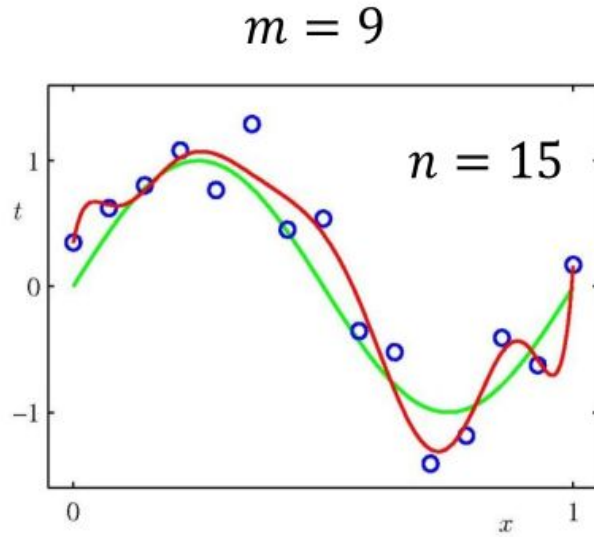
Model complexity

Example:

- Polynomials with larger m are becoming increasingly the random noise on the target values.



Number of training data and overfitting



Evaluation and Model selection

► **Evaluation:**

- We need to measure how well the learned function can predicts the target for unseen examples

► **Model selection:**

- Most of the time we need to select among a set of models
 - Example: polynomials with different degree m
- and thus we need to evaluate these models first

Avoiding overfitting

Determine a suitable value for model complexity

- ▶ **Simple hold-out method**
- ▶ **Cross-validation**

Regularization (Occam's Razor)

- ▶ Explicit preference towards simple models
- ▶ Penalize for the model complexity in the objective function

Simple hold-out model selection

- ▶ Steps:

- ▶ Divide training data into training and validation set v_set
- ▶ Use only the training set to train a set of models
- ▶ Evaluate each learned model on the validation set
 - ▶ $J_v(\mathbf{w}) = \frac{1}{|v_set|} \sum_{i \in v_set} \left(y^{(i)} - f(\mathbf{x}^{(i)}; \mathbf{w}) \right)^2$
- ▶ Choose the best model based on the validation set error

- ▶ Usually, too wasteful of valuable training data

- ▶ Training data may be limited.
- ▶ On the other hand, small validation set give a relatively noisy estimate of performance.

Simple hold-out

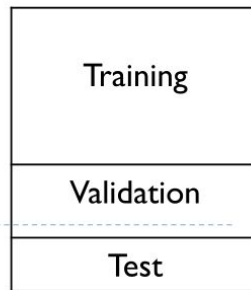
Simple hold-out chooses the model that minimizes error on validation set.

$J_v(\hat{\mathbf{w}})$ is likely to be an optimistic estimate of generalization error.

- ▶ extra parameter (e.g., degree of polynomial) is fit to this set.

Estimate generalization error for the test set

- ▶ performance of the selected model is finally evaluated on the test set



Cross validation (CV) : Evaluation

k -fold cross-validation steps:

- ▶ Shuffle the dataset and randomly partition training data into k groups of approximately equal size
- ▶ for $i = 1$ to k
 - ▶ Choose the i -th group as the held-out validation group
 - ▶ Train the model on all but the i -th group of data
 - ▶ Evaluate the model on the held-out group
- ▶ Performance scores of the model from k runs are **averaged**.
 - ▶ The average error rate can be considered as an estimation of the true performance.



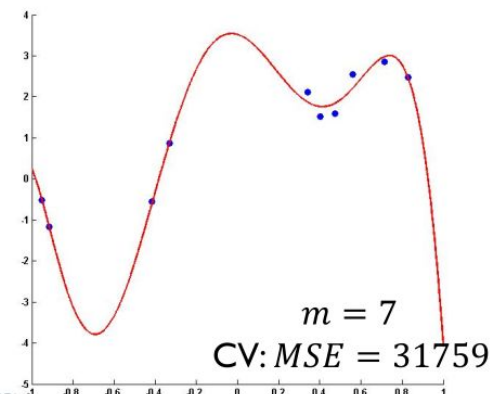
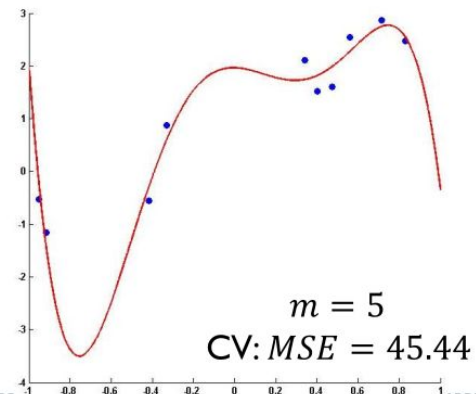
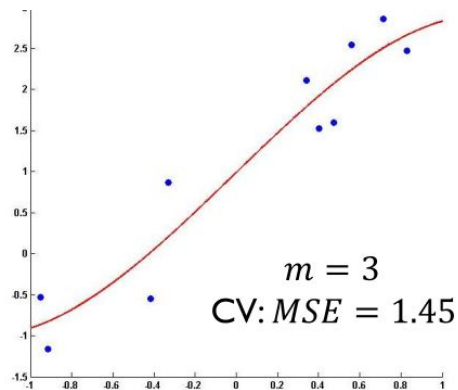
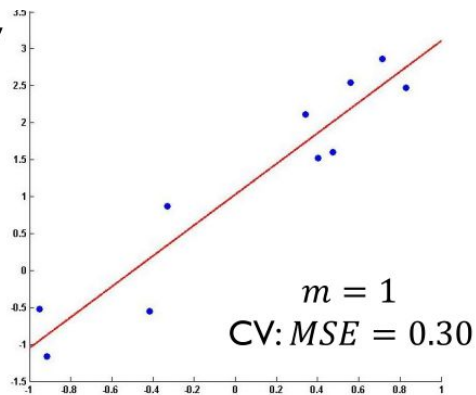
Cross validation (CV) : Evaluation

For each model we first find the average error find by CV.

The model with **the best average performance** is selected.

Cross-validation: polynomial regression example

- ▶ 5-fold CV
- ▶ 100 runs
- ▶ average



Reference

Soleimani, M . *Machine Learning Course Lecture Slides*.

C. Bishop, “Pattern Recognition and Machine Learning”, Chapter 1.1,1.3